

Othmane Echchabi

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EDUCATION

McGill University & Mila – Quebec AI Institute — *M.Sc. in Computer Science (GPA: 4.0/4.0)* Montréal, QC | May 2027
Duke University / Duke Kunshan University — *B.S. in Data Science* Durham, NC / Suzhou, China | May 2025
University Carlos III Madrid — *Study Abroad – Data Engineering* Madrid, Spain | May 2022

RESEARCH EXPERIENCE

Graduate Research Assistant — *Rolnick Lab, McGill University & Mila* Aug. 2025 – Present | Montréal, QC

- Leading a tree-mortality classification project with the Smithsonian Institution on Barro Colorado Island (BCI), Panama.

Machine Learning Research Intern — *College of Computing, UM6P* Jul. 2025 – Dec. 2025 | Benguerir, Morocco

- Integrated a 10-year crop-yield dataset with N/P/K fertilizer inputs across 150+ countries for downstream modeling.
- Built a transformer-based recommendation model and a RAG assistant for agronomy; reduced query latency to ~2s.

Undergraduate Research Assistant — *Sustainability Assessment Group, DKU* Jun. 2024 – Aug. 2025 | Suzhou, China

- Built a 300k satellite image piped-water access dataset; achieved SotA results with SSL models; validated against UN data; ran large-scale inference across 50/54 African countries using 1M+ Sentinel images and population-density covariates.
- Built SpeedTransformer for Geolife transport-mode prediction (96% Top-1, +3% vs. baseline), and prototyped CarbonClever, a phone application, to deliver personalized guidance for reducing personal CO₂ emissions.

PUBLICATIONS & PREPRINTS

1. Zhang, Y.*; **Echchabi, O.***; Feng, T.; Zhang, W.; Liao, H.-K.; Chang, C. *Detecting Transportation Mode Using Dense Smartphone GPS Trajectories and Transformer Models*. Accepted at *IJGIS* (2026).
2. **Echchabi, O.***; Lahlou, A.*; Talty, N.; Manto, J. M.; Tongshu, Z.; Lam, K. L. *Monitoring Access to Piped Water and Sanitation Infrastructure in Africa at Disaggregated Scales Using Satellite Imagery and Self-Supervised Learning*. Accepted at the *ML4RS Workshop, ICLR* (2026).

POSTERS & PRESENTATIONS

- Monitoring *Spartina alterniflora* Using Self-Supervised Learning. *Duke Climate+ Symposium* (2024).
- Assessing Climate Change Risk of Rural Coastal Plains. *Duke Climate+ Symposium* (2023).
- Scalable Water and Sanitation Monitoring Across Africa via AI-Driven Satellite and Survey Data. *AGU* (2025).

GRANTS & AWARDS

- McGill CS Graduate Award & Registration Bonus — \$35,000/year
- DKU Grants: Thesis Research Grant (SWRG & SELF): \$2,500 — DKU Innovation Incubator: \$3,000

TEACHING EXPERIENCE

- **McGill University** — TA — COMP 303;
- **FADI Academy - Online College Prep Center** — Partner & Math Tutor
- **Duke Kunshan University** — TA — COMPSCI 101, COMPSCI 201, MATH 105, MATH 201, MATH 202, MATH 317;

SERVICE & LEADERSHIP

- **Mila - AI Quebec Institute** — Mila Lab Representative (2025–Present)
- **McGill University** — Project Development VP (Board Member), Muslims in Tech (2025–Present)
- **Duke Kunshan University** — Resident Assistant; Soccer Varsity Co-Captain; CS Club Software Tech Lead
- **FIFA World Cup 2022** — Team Leader, Spectator Services Volunteer

MISCELLANEOUS

Technical Skills : Python, PyTorch, Docker, Kubernetes, OpenCV, Linux, Git, TensorFlow, Keras, scikit-learn, Java

Spoken Languages : Fluent in Arabic, French, English; Intermediate Spanish and Chinese

Interests : Competitive Soccer, Weightlifting, Cooking, Traveling.